सहायक प्राध्यापक (दंतपरिवेष्टनशास्त्र, दंतविकृती व अणुजीवशास्त्र, मुखशल्यचिकित्साशास्त्र)
शासकीय दंत महाविद्यालय व रुग्णालय, वैद्यकिय शिक्षण व संशोधन संचालनालय, गट-ब
वैद्यकिय शिक्षण व औषधी द्रव्ये विभाग
Assistant Professor (Periodontology, Oral Pathology and Microbiology, Oral and Maxillofacial Surgery) Government Dental College and Hospital, Directorate of Medical Education and Research, Group-B
Medical Education and Drugs Department

परीक्षेचे टप्पे:- १) लेखी परीक्षा- २०० गुण **२)** मुलाखत - ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक (१११०)	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विभाग-अ विषयाशी/विभागाशी संबंधित घटक	इंग्रजी	५०	१००	एक तास	पदवी	वस्तुनिष्ठ बहुपर्यायी
विभाग-ब दंतपरिवेष्टनशास्त्र		५०	१००		पदव्युत्तर पदवी	
विभाग-क दंतविकृती व अणुजीवशास्त्र		५०	१००			
विभाग-ड मुखशल्यचिकित्साशास्त्र		५०	१००			

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

क) दंतपरिवेष्टनशास्त्र विषयाकरीता विभाग-अ व विभाग-ब, दंतविकृती व अणुजीवशास्त्र विषयाकरीता विभाग-अ व विभाग-क व मुखशल्यचिकित्साशास्त्र विषयाकरीता विभाग-अ व विभाग-ड याप्रमाणे अभ्यासक्रम विहित करण्यात आला आहे.

-: अभ्यासक्रम :-

विषयाशी/विभागाशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

Sr. No.	Topics and sub-topics
	SECTION A
1.	**HUMAN ANATOMY, EMBRYOLOGY, HISTOLOGY AND MEDICAL GENETICS** I. INTRODUCTION TO : 1. Anatomical terms. 2. Skin, superficial fascia and deep fascia 3. Cardiovascular system, portal system collateral circulation and arteries. 4. Lymphatic system, regional lymph nodes 5. Osteology - Including ossification and growth of bones 6. Myology – Including types of muscle tissue and innervation.

7. Syndesmology – Including classification of Joints.
8. Nervous system

II. HEAD and NECK:

1. Scalp, face and temple, lacrimal apparatus
2. Neck - Deep fascia of neck, posterior triangle, suboccipital triangle, anterior triangle, anterior median region of the neck, deep structures in the neck.
3. Cranial cavity - Meninges, parts of brain, ventricles of brain, dural venous sinuses, cranial nerves attached to the brain, pituitary gland.
4. Cranial nerves - III, IV, V, VI, VII, IX,XII in detail.
5. Orbital cavity – Muscles of the eye ball, supports of the eye ball, nerves and vessels in the orbit.
6. Parotid gland.
7. Temporo mandibular joint, muscles of mastication, infratemporal fossa, pterygo - palatine fossa.
8. Submandibular region
9. Walls of the nasal cavity, paranasal air sinuses
10. Palate
11. Oral cavity, Tongue
12. Pharynx (palatine tonsil and the auditory tube) Larynx. OSTEOLOGY – Foetal skull, adult skull, individual bones of the skull, hyoid bone and cervical vertebrae

III. THORAX : Demonstration on a dissected specimen of

1. Thoracic wall
2. Heart chambers
3. Coronary arteries
4. Pericardium
5. Lungs – surfaces ; pleural cavity
6. Diaphragm

IV. ABDOMEN : Demonstration on a dissected specimen of

1. Peritoneal cavity
2. Organs in the abdominal and pelvic cavity.

V. CLINICAL PROCEDURES :

a) Intramuscular injections: Demonstration on a dissected specimen and on a living person of the following sites of injection.

1. Deltoid muscle and its relation to the axillary nerve and radial nerve.
2. Gluteal region and the relation of the sciatic nerve.
3. Vastus lateralis muscle.

b) Intravenous injections and venesection: Demonstration of veins in the dissected specimen andon a living person.

1. Median cubital vein
2. Cephalic vein
3. Basilic vein
4. Long saphenous vein

c) Arterial pulsations: Demonstration of arteries on a dissected specimen and feeling ofpulsation of the following arteries on a living person.

1. Superficial temporal
2. Facial
3. Carotid
4. Axillary
5. Brachial
6. Radial
7. Ulnar
8. Femoral
9. Popliteal

10. Dorsalispedis Lumbar bar puncture: Demonstration on a dissected specimen of the spinal cord, cauda equina and epidural space and the inter vertebral space between L4 and L5 .

VI. EMBRYOLOGY :

Oogenesis, Spermatogenesis, Fertilisation, Placenta, Primitive streak, Neural crest, Bilaminar and trilaminar embryonic disc, Intra embryonic mesoderm - formation and fate, notochord formation and fate, Pharyngeal arches, pouches and clefts, Development of face, tongue, palate, thyroid gland, pituitary gland, salivary glands, and anomalies in their development, Tooth development in brief.

VII. HISTOLOGY :

The Cell :

Basic tissues - Epithelium, Connective tissue including cartilage and bone, Muscle Tissue, Nervous tissue : Peripheral nerve, optic nerve, sensory ganglion, motor ganglion, Skin Classification of Glands

Salivary glands (serous, mucous and mixed gland), Blood vessels, Lymphoid tissue Tooth, lip, tongue, hard palate, oesophagus, stomach, ,duodenum ,ileum, colon, vermiform appendix Liver, Pancreas, Lung, Trachea ,Epiglottis, Thyroid gland , para thyroid gland , supra renal gland and pituitary gland, Kidney, Ureter, Urninary bladder, Ovary and testis.

VIII. MEDICAL GENETICS :

Mitosis, meiosis, Chromosomes, gene structure, Mendelism, modes of inheritance

2. **HUMAN PHYSIOLOGY**

1. GENERAL PHYSIOLOGY

1. Homeostasis: Basic concept, Feed back mechanisms
2. Structure of cell membrane, transport across cell membrane
3. Membrane potentials

2. BLOOD:

Composition and functions of blood.

Specific gravity, Packed cell volume, factors affecting and methods of determination. Plasma proteins - Types, concentration, functions and variations.

Erythrocyte - Morphology, functions and variations. Erythropoiesis and factors affecting erythropoiesis. ESR- Methods of estimation, factors affecting, variations and significance.

Haemoglobin - Normal concentration, method of determination and variation in concentration. Blood Indices - MCV, MCH, MCHC - definition, normal values, variation.

- Definition, classification, life span of RBC's destruction of RBC's , formation and fate of bilepigments, Jaundice - types.

Leucocytes : Classification, number, percentage, distribution morphology, properties, functions and variation. Role of lymphocytes in immunity, leucopoiesis life span and fate of leucocytes.

Thrombocytes - Morphology, , number, variations, function and thrombopoiesis.

Haemostasis - Role of vasoconstriction, platelet plug formation in haemostasis, coagulation factors, intrinsic and extrinsic pathways of coagulation, clot retraction. Tests of haemostatic function, platelet count, clotting time, bleeding time, prothrombin time - normal values, method and variations. Anticoagulants - mechanism of action. Bleeding disorders.

Blood groups: ABO and Rh system, method of determination, importance, indications and dangers of bloodtransfusion, blood substitutes.

Blood volume: Normal values, variations.

Body fluids : distribution of total body water, intracellular and extracellular

compartments, major anions and cations in intra and extra cellular fluid.
Tissue fluids and lymph : Formation of tissue fluid, composition, circulation and functions of lymph. Oedema - causes.
Functions of reticulo endotrelial system.

3. MUSCLE AND NERVE

Classification of nerves, structure of skeletal muscle - Molecular mechanism of muscle contraction, neuromuscular transmission. Properties of skeletal muscle. Structure and properties of cardiac muscle and smooth muscle.

4. DIGESTIVE SYSTEM :

Introduction to digestion : General structure of G.I. tract, Innervation.
Salivary glands: Structure of salivary glands, composition, regulation of secretion and functions of saliva. Stomach: Composition and functions of gastric juice, mechanism and regulation of gastric secretion.
Exocrine Pancreas - Structure, composition of pancreatic juice, functions of each component, regulation of pancreatic secretion.
Liver : structure , composition of bile, functions of bile, regulation of secretion – Gall bladder : structure, functions.
Small intestine - Composition, functions and regulation of secretion of intestinal juice. Large intestine - Functions.
Motor functions of GIT: Mastication, deglutition, gastric filling and emptying, movements of small and large intestine, defecation.

5. EXCRETORY SYSTEM :

Structure and functions of kidney, functional unit of kidney and functions of different parts. Juxta glomerular apparatus, renal blood flow.
Formation of Urine : Glomerular filteration rate - definition, determination , normal values, factors influencing G.F.R. Tubular reabsorption - Reabsorption of sodium, glucose, water and other substances. Tubular secretion - secretion of urea, hydrogen and other substances.
Mechanism of concentration and dilution of urine. Role of kidney in the regulation of pH of the blood.
Micturition : anatomy and innervation of Urinary bladder, mechanism of miturition and abonrmalities.

6. BODY TEMPERATURE AND FUNCTIONS OF SKIN

7. ENDOCRINOLOGY

General endocrinology - Enumeration of endocrine glands and hormones - General functions of endocrine system, chemistry, mechanism of secretion, transport, metabolism, regulation of secretion of hormones. Hormones of anterior pituitary and their actions, hypothamic regulation of anterior pituitary function. Disorders of secretion of anterior pituitary hormones.
Posterior pituitary : Functions, regulation and disorders of secretion.
Thyroid: Histology, synthesis, secretion and transport of hormones, actions of hormones, regulation of secretion and disorders, Thyroid function tests.
Adrenal cortex and Medulla -synthesis, secretion, action, metabolism, regulation of secretion of hormones and disorders.
Other hormones - Angiotensin, A.N.F.

8. REPRODUCTION

Sex differentiation, Physiological anatomy of male and female sex organs,
Female reproductive system : Menstrual cycle, functions of ovary, actions of oestrogen and Progesterone, control of secretion of ovarian hormones, tests for ovulation, fertilisation, implantation, maternal changes during pregnancy,

pregnancy tests and parturition.
Lactation, composition of milk, factors controlling lactation, milk ejection, reflex, Male reproductive system :spermatogenesis, semen and contraception.

9. CARDIO VASCULAR SYSTEM

Functional anatomy and innervation of heart Properties of cardiac muscle Origin and propagation of cardiac impulse and heart block.
Electrocardiogram - Normal electrocardiogram.Two changes in ECG in myocardial infarction.
Cardiac cycle - Phases, Pressure changes in atria, ventricles and aorta. Volume changes in ventricles. Jugular venous pulse, arterial pulse.
Heart sounds: Mention of murmurs.
Heart rate: Normal value, variation and regulation.
Cardiac output: Definition, normal values, one method of determination, variation, factors affectingheart rate and stroke volume.
Arterial blood pressure: Definition, normal values and variations, determinants, regulation and measurementof blood pressure.
Coronary circulation.
Cardio vascular homeostasis - Exercise and posture.

10. RESPIRATORY SYSTEM

Physiology of Respiration : External and internal respiration. Functional anatomy of respiratory passage and lungs.
Respiratory movements: Muscles of respiration, Mechanism of inflation and deflation of lungs. Intra pleural and intra pulmonary pressures and their changes during the phases of respiration. Mechanics of breathing - surfactant, compliance and work of breathing.
Spirometry: Lung volumes and capacities definition, normal values, significance, factors affecting vital capacity, variations in vital capacity, FEV and its variations.
Pulmonary ventilation - alveolar ventilation and dead space – ventilation. Composition of inspired air, alveolar air and expired air.
Exchange of gases: Diffusing capacity, factors affecting it. Transport of Oxygen and carbon dioxide in the blood.
Regulation of respiration – neural and chemical. Hypoxia, cyanosis, dyspnoea, periodic breathing.Artificial respiration, pulmonary function tests.

11. CENTRAL NERVOUS SYSTEM

1. Organisation of central nervous system
2. Neuronal organisation at spinal cord level
3. Synapse receptors, reflexes, sensations and tracts
4. Physiology of pain
5. Functions of cerebellum, thalamus, hypothalamus and cerebral cortex.
6. Formation and functions of CSF
7. Autonomic nervous system

12. SPECIAL SENSES

Fundamental knowledge of vision, hearing, taste and smell.

3. **<u>BIOCHEMISTRY</u>**

1. CHEMISTRY OF BIOORGANIC MOLECULES

Carbohydrates: Definition, biological importance and classification. Monosaccharides - Isomerism, anomerism. Sugar derivatives, Disaccharides. Polysaccharides. Structures of starch and glycogen.
Lipids : Definition, biological importance and classification. Fats and fatty acids. Introduction to compound lipids. Hydrophobic and hydrophilic groups.

Cholesterol. Bile salts. Micelle. Bimolecular leaflet.

Proteins: Biological importance. Aminoacids: Classification. Introduction to peptides. Proteins : Simple and conjugated; globular and fibrous. Charge properties. Buffer action. Introduction to protein conformation . Denaturation.
Nucleic acids: Building units . Nucleotides. Outline structure of DNA and RNA. High energy compounds: ATP, Phosphorylamidines, Thiolesters, Enol phosphates.

2. MACRONUTERIENTS AND DIGESTION

Energy needs: Basal metabolic rate. Dietary carbohydrates, fibres. Dietary lipids, essential fatty acids. Nitrogen balance. Essential amino acids. Protein quality and requirement (methods for evaluation of protein quality to be excluded). Protein calorie malnutrition. Balanced diet.
Enzymatic hydrolysis of dietary carbohydrates. Mechanism of uptake of monosaccharides. Digestion and absorption of triacylglycerols. Enzymatic hydrolysis of dietary proteins and uptake of amino acids.

3. MICRONUTRIENTS

Vitamins: Definition, classification, daily requirement, sources and deficiency symptoms. Brief account of water-soluble vitamins with biochemical functions. Vitamins A functions including visual process. Vitamin D and its role in calcium metabolism. Vitamin E. Vitamin K and gamma carboxylation. Introduction to antivitamins and hypervitaminosis.
Minerals :Classification, daily requirement. Calcium and phosphate: sources, uptake, excretion, function. Serum calcium regulation. Iron: sources, uptake and transport.
Heme and nonheme iron functions; deficiency. Iodine: Brief introduction to thyroxine synthesis. General functions of thyroxine. Fluoride: function, deficiency and excess. Indications of role of other minerals.

4. ENERGY METABOLISM

Overview: Outlines of glycolysis, pyruvate oxidation and citric acid cycle. Beta oxidation of fatty acids. Electron transport chain and oxidative phosphyorylation. Ketone body formation and utilisation. Introduction to glycogenesis, glycogenolysis, fatty acid synthesis, lipogenesis and lipolysis. Gluconeogenesis. Lactate metabolism .Protein utilisation for energy. Glucogenic and ketogenic amino acids. Integration of metabolism.

5. SPECIAL ASPECTS OF METABOLISM

Importance of pentose phosphate pathway. Formation of glucuronic acid. Outlines of cholesterol synthesis and breakdown. Ammonia metabolism. Urea formation. Phosphocreatine formation. Transmethylation. Amines. Introduction to other functions of amino acids including one carbon transfer. Detoxication : Typical reactions. Examples of toxic compounds. Oxygen toxicity

6. BIOCHEMICAL GENETICS AND PROTEIN SYNTHESIS

Introduction to nucleotides; formation and degradation. DNA as genetic material. Introduction to replication and transcription. Forms and functions of RNA. Genetic code and mutation. Outline of translation process. Antimetabolites and antibiotics interfering in replication, transcription and translation. Introduction to cancer, viruses and oncogenes.

7. ENZYME AND METABOLIC REGULATION

Enzymes: Definition, classification, specificity and active site. Cofactors. Effect of pH, temperature and substrate concentration. Introduction to enzyme inhibitors, proenzymes and isoenzymes. Introduction to allosteric regulation, covalent modification and regulation by induction/repression.

Overview of hormones. Introduction to second messengers, cyclic AMP, calcium ion, inositol triphosphate. Mechanism of action of steroid hormones, epinephrine, glucagon and insulin in brief. Acid base regulation. Electrolyte balance.

8. STRUCTURAL COMPONENTS AND BLOOD PROTEINS

Connective tissue: Collagen and elastin. Glycosaminoglycans. Bone structure. Structure of membranes. Membrane associated processes in brief. Exocytosis and endocytosis. Introduction to cytoskeleton. Myofibril and muscle contraction in brief.

Haemoglobin: functions. Introduction to heme synthesis and degradation. Plasma proteins: classification and separation. Functions of albumin. A brief account of immunoglobulins. Plasma lipoproteins: Formation, function and turnover.

9. MEDICAL BIOCHEMISTRY

Regulation of blood glucose. Diabetes mellitus and related disorders. Evaluation of glycemic status. Hyperthyroidism and hypothyroidism: Biochemical evaluation. Hyperlipoproteinemias and atherosclerosis, Approaches to treatment. Jaundice: Classification and evaluation. Liver function tests: Plasma protein pattern, serum enzymes levels. Brief introduction to kidney function tests and gastric function tests. Acid base imbalance. Electrolyte imbalance: evaluation. Gout. Examples of genetic disorders including lysosomal storage disorders, glycogen storage disorders, glucose 6- phosphate dehydrogenase deficiency, hemoglobinopathies, inborn errors of amino acid metabolism and muscular dystrophy (one or two examples with biochemical basis will be adequate). Serum enzymes in diagnosis.

4. **GENERAL PATHOLOGY**

1. Introduction to PathologyTerminologies
The cell in health
The normal cell structureThe cellular functions
2. Etiology and Pathogenesis of DiseaseCell Injury
Types – congenital
Acquired
Mainly Acquired causes of disease
(Hypoxic injury, chemical injury, physical injury, immunological injury)
3. Degenerations
Amyloidosis
Fatty change
Cloudy
swelling
Hyaline change, mucoid degeneration
4. Cell death and NecrosisApoptasis
Def, causes, features and types of necrosisGangrene - Dry, wet, gas
Pathological Calcifications (Dystrophic and metastatic)
5. Inflammation
Definition, causes types, and features
Acute inflammation
a. The vascular response
b. The cellular response
c. Chemical mediators
d. The inflammatory cells
e. Fate
Chronic inflammation
Granulomations inflammation
6. Healing

- Regeneration
- Repair

a. Mechanisms
b. Healing by primary intention
c. Healing by secondary intention
d. Fracture healing
e. Factors influencing healing process
f. Complications

7. Tuberculosis

- Epidemiology
- Pathogenesis (Formation of tubercle)
- Pathological features of Primary and secondary TB
- Complications and Fate

8. Syphilis

- Epidemiology
- Types and stages of syphilis

Pathological features
- Diagnostic criterias
- Oral lesions

9. Typhoid

Epidemiology
Pathogenesis
Pathological features
Diagnostic criterias

10. Thrombosis

- Definition, Pathophysiology
- Formation, complications and Fate of a thrombus

11. Embolism

- Definition
- Types
- Effects

12. Ischaemia and Infraction

- Definition, etiology, types
- Infraction of various organs

13. Derangements of body fluids

Oedema – pathogenesis
Different types

14. Disorders of circulation

- Hyperaemia
- Shock

15. Nutritional Disorders

- Common Vitamin Deficiencies

16. Immunological mechanisms in disease

Humoral and cellular immunity
Hypersensitivity and autoimmunity

17. AIDS and Hepatitis.

18. Hypertension

Definition, classification
Pathophysiology
Effects in various organs

19. Diabetes Mellitus

-Def, Classification, Pathogenesis, Pathology in different organs

20. Adaptive disorders of growth

-Atrophy and Hypertrophy, Hyperplasia, Metaplasia and Dysplasia

21. General Aspects of neoplasia
a. Definition, terminology, classification
b. Differences between benign and malignant neoplasms
c. The neoplastic cell
d. Metastasis
e. Etiology and pathogenesis of neoplasia, Carcinogenesis
f. Tumour biology
g. Oncogenes and anti-oncogenes
h. Diagnosis
i. Precancerous lesions
j.Common specific tumours, Sq papilloma and Ca, Basal cell Ca, Adenoma and Adenoca, FibromaandFibrosarcoma, Lipoma and liposarcoma
A.Systemic Pathology –
22 Anaemias
- Iron Deficiency anaemia, Megaloblastic anaemia

23.Leukaemias
- Acute and chronic leukaemias, Diagnosis and clinical features

24. Diseases of Lymph nodes
- Hodgkin's disease, Non Hodgkins lymphoma, Metastatic carcinoma

25. Diseases of oral cavity
Lichen planus, Stomatitis, Leukoplakia, Sq cell Ca, Dental caries, Dentigerious cyst,Ameloblastoma

26. Diseases of salivary glands
Normal structure, Sialadenitis, Tumours

27. Common diseases of Bones
Osteomyelitis, Metabolic bone diseases, Bone Tumours, Osteosarcoma, Osteocalstoma, Giant cell Tumour, Ewing's sarcoma, Fibrous dysplasia, Aneurysmal bone cyst

28. Diseases of Cardiovascular system
Cardiac failuare
Congenital heart disease – ASD,
VSD,PDAFallot's Tetrology

Infective Endocarditis
Atherosclerosis
Ischaemic heart Disease

29. Haemorrhagic DisordersCoagulation cascade Coagulation disorders
Platelet funtion
Platelet disorders

5. **MICROBIOLOGY**

1. History, Introduction, Scope, Aims and Objectives.
2. Morphology and Physiology of bacteria.
3. Detail account of Sterlisation and Disinfection.
4. Brief account of Culture media and Culture techniques.
5. Basic knowledge of selection, collection, transport, processing of clinical Specimens and identification of bacteria.
6. Bacterial Genetics and Drug Resistance in bacteria.

B. IMMUNOLOGY:

1. Infection - Definition, Classification, Source, Mode of transmission and types of Infectious disease.
2. Immunity
3. Structure and functions of Immune system
4. The Complement System

	5. Antigen 6. Immunoglobulins - Antibodies - General structure and the role played in defense mechanism of the body. 7. Immune response 8. Antigen - Antibody reactions - with reference to clinical utility. 9. Immuno deficiency disorders - a brief knowledge of various types of immuno deficiency disorders - A sound knowledge of immuno deficiency disorders relevant to dentistry. 10. Hypersensitivity reactions 11. Autoimmune disorders - Basic knowledge of various types - sound knowledge of autoimmune disorders of oral cavity and related structures. 12. Immunology of Transplantation and Malignancy 13. Immunehaematology C. SYSTEMATIC BACTERIOLOGY: 1. Pyogenic cocci - Staphylococcus, Streptococcus, Pneumococcus, Gonococcus, Meningococcus – brief account of each coccus - detailed account of mode of spread, laboratory diagnosis, Chemo therapy and prevention - Detailed account of Cariogenic Streptococci. 2. Corynebacterium diphtheriae - mode of spread, important clinical feature, Laboratory diagnosis, Chemotherapy and Active immunisation. 3. Mycobacteria - Tuberculosis and Leprosy 4. Clostridium - Gas gangrene, food poisoning and tetanus. 5. Non-sporing Anaerobes - in brief about classification and morphology, in detail about dental pathogens - mechanism of disease production and prevention. 6. Spirochaetes - Treponema pallidum - detailed account of Oral Lesions of syphilis, Borrelia vincentii. 7. Actinomycetes. D. VIROLOGY: 1. Introduction 2. General properties, cultivation, host - virus interaction with special reference to Interferon. 3. Brief account of Laboratory diagnosis, Chemotherapy and immuno prophylaxis in general. 4. A few viruses of relevance to dentistry. • Herpes Virus • Hepatitis B Virus - brief about other types • Human Immunodeficiency Virus (HIV) • Mumps Virus • Brief - Measles and Rubella Virus 5. Bacteriophage - structure and Significance E. MYCOLOGY 1. Brief Introduction 2. Candidosis - in detail 3. Briefly on oral lesions of systemic mycoses. F. PARASITOLOGY: 1. Brief introduction - protozoans and helminths 2. Brief knowledge about the mode of transmission and prevention of commonly seen parasitic infection in the region.
6.	**GENERAL AND DENTAL PHARMACOLOGY AND THERAPEUTICS** 1. General principles of pharmacology; sources and nature of drugs dosage forms; prescription writing; pharmacokinetics (absorption, distribution, metabolism and

excretion of drugs), mode of action of drugs, combined effects of drugs, receptor mechanism of drug action, factors modifying drug response, adverse drug reactions; drug interactions, Implications of General Principles in clinical dentistry.

2. CNS drugs; General anaesthetics, hypnotics, analgescis psychotropic drugs, anti – epileptics, muscle relaxants, local anaesthetics, Implications of these drugs in clinical dentistry.
3. Autonomic drugs; sympathomimetics, antiadrenergic drugs parasympothomimetics and parasympatholytics, Implications of Autonomic drugs in clinical dentistry.
4. Cardiovascular drugs; Cardiac stimulants ; antihypertensive drugs, vasopressor agents, treatment of shock, Antianginal agents and diuretics, Implications of these drugs in clinical dentistry.
5. Autocoids:
   Histamine, antihistamines, prostaglandins, leukotriens and bronchodilators, Implications ofAutocoids in clinical dentistry.
6. Drugs acting on blood : coagulants and anticoagulants, hematinics, Implications of these drugs in clinical dentistry.
7. G.I.T. Drugs, Purgatives, anti-diarrhoeal, antacids, anti-emetics, Implications of these drugs in clinical dentistry.
8. Endocrines; Emphasis on treatment of diabetes and glucocorticoids, thyroid and antithyroid agents, drugs affecting calcium balance and anabolic steroids, Implications of these drugs in clinical dentistry.
9. Chemotherapy: Antimicrobial agents (against bacteria, anaerobic infections, fungi, virus and broad spectrum). Infection management in dentistry. Phamacotherapy of Tuberculosis, leprosy and chemotherapy of malignancy in general. Implications of Chemotherpy in clinical dentistry.
10. Vitamins : Water soluble vitamins, Vit. D, Vit.K. and Vit. E, Implications of Vitamins in clinical dentistry.
11. Pharmacotherapy of emergencies in dental office and emergency drugs tray Implications of Pharmacotherapy in clinical dentistry.
12. Chealating agents – BAL,EDTA and desferrioxamine,

DENTAL PHARMACOLOGY

1. Anti - septics, astrigents, obtundents, mummifying agents, bleaching agents, styptics, disclosingagents, dentifrices, mouth washes, caries and fluorides.
2. Pharmacotherapy of common oral conditions in dentistry.

7. GENERAL MEDICINE

CORE TOPICS
(Must Know)

1. Aims of medicine Definitions of signs, symptoms, diagnosis, differential diagnosis treatment and prognosis.

2. Infections.
Enteric fever, AIDS, herpes simplex, herpes zoster, syphilis diphtheria.

Infectious mononucleosis mumps, measles, rubella, malaria.

3. G.I.T.
Stomatitis, gingival hyperplasia, dysphagia, acid peptic disease, jaundice, acute and chronic hepatitis, cirrhosis of liver ascites.

Diarrhea Dysentery Amoebiasis Malabsorhtion

4. CVS
Acute rheumatic fever rheumatic valvular heart disease, hypertension, ischemic heart disease, infective endocarditis, common arrhythmias,

	congenital heart disease, congestive cardiac failure. 5. RS Pneumonia, COPD, Pulmonary TB, Bronehial asthma	Lung Abscess Pleural effusion Pneumothorax Bronchiectasis Lung cancers.
	6. Hematology Anemias, bleeding and clotting disorders, leukemias, lymphomas, agranulocytosis, splenomegaly, oral manifestations of hematologic disorders, generalized Lymphadenopathy.	
	7. Renal System Acute nephritis Nephrotic syndrome	Renal failure
	8. Nutrition Avitaminosis	Balanced diet PEM Avitaminosis
	9. CNS Facial palsy, facial pain including trigeminal neuralgia, epilepsy, headache including migraine.	Meningitis Examination of comatose patient Examination of cranial nerves.
	10. Endocrines Diabetes Mellitus Acromegaly, Hypothyroidism, Thyrotoxicosis, Calcium metabolism and parathyroids.	Addison's disease, Cushing's syndrome.
	11. Critical care Syncope, cardiac arrest, CPR, shock	
8.	**GENERAL SURGERY** 1. HISTORY OF SURGERY: The development of surgery as a speciality over the years, will give the students an opportunity to know the contributions made by various scientists, teachers and investigators. It will also enable the student to understand the relations of various specialities in the practice of modern surgery. 2. GENERAL PRINCIPLES OF SURGERY: Introduction to various aspects of surgical principles as related to orodental diseases. Classification of diseases in general. This will help the student to understand the various diseases, their relevance to routine dental practice. 3. WOUNDS: Their classification, wound healing, repair, treatment of wounds, medico-legal aspects ofaccidental wounds and complications of wounds. 4. INFLAMMATION: Of soft and hard tissues. Causes of inflammation, varieties, treatment and sequelae. 5. INFECTIONS: Acute and chronic abscess skin infections, cellulitis, carbuncle, and erysepelas. Specific infections such as tetanus, gangrene, syphilis, gonorrhoea, tuberculosis, Actinomycosis, Vincents angina, cancrum oris. Pyaemia, toxaemia and septicaemia. 6. TRNSMISSABLE VIRAL INFECTIONS: HIV and Hepatitis B with special reference to their prevention and precautions to be taken in treating patients in a carrier state. 7. SHOCK AND HAEMORRHAGE: Classification, causes, clinical features and management of various types of shock.	

Syncope, Circulatory collapse. Haemorrhage – different types, causes, clinical features and management. Blood groups, blood transfusion, precautions and complications of blood and their products. Hemophilia's, their transmission, clinical features and management especially in relation to minor dental procedures.

8. TUMOURS, ULCERS, CYSTS, SINUS AND FISTULAE:
   Classification, clinical examination and treatment principles in various types of benign and malignant tumours, ulcers, cysts, sinus and fistulae.

9. DISEASES OF LYMPHATIC SYSTEM:
   Especially those occurring in head and neck region. Special emphasis on identifying diseases such as tubercular infection, lymphomas, leukaemias, metastatic lymph node diseases.

10. DISEASES OF THE ORAL CAVITY:
    Infective and malignant diseases of the oral cavity and oropharynx including salivary glands with special emphasis on preventive aspects of premalignant and malignant diseases of the oral cavity.

11. DISEASES OF LARYNX, NASOPHARYNX:
    Infections and tumours affecting these sites. Indications, procedure and complications of tracheostmy.

12. NERVOUS SYSTEM:
    Surgical problems associated with nervous system with special reference to the principles of peripheral nerve injuries, their regeneration and principles of treatment. Detailed description of afflictions of facial nerve and its management. Trigeminal neuralgia, its presentation and treatment.

13. FRACTURES:
    General principles of fractures, clinical presentation and treatment with additional reference to newer methods of fracture treatment. Special emphasis on fracture healing and rehabilitation.

14. PRINCIPLES OF OPERATIVE SURGERY:
    Principles as applicable to minor surgical procedures including detailed description of asepsis, antiseptics, sterilisation, principles of anaesthesia and principles of tissue replacement. Knowledge of sutures, drains, diathermy, cryosurgery and use of Laser in surgery.

15. ANOMOLIES OF DEVELOPMENT OF FACE:
    Surgical anatomy and development of face. Cleft lip and cleft palate—principles of management.

16. DISEASES OF THYROID AND PARATHYROID:
    Surgical anatomy, pathogenesis, clinical features and management of dysfunction of thyroid and parathyroid glands. Malignant diseases of the thyroid—classification, clinical features and management.

17. SWELLINGS OF THE JAW:
    Differential diagnosis and management of different types of swellings of the jaw.

18. BIOPSY:
    Different types of biopsies routinely used in surgical practice.
    Skills to be developed by the end of teaching is to examine a routine swelling, ulcer and other related diseases and to perform minor surgical procedures such as draining an abscess, taking abiopsy etc.

9. RESEARCH METHODOLOGY AND DENTAL EDUCATION TECHNOLOGY

1. Health Information: - Basic knowledge of Computers, MS Office, Window 2000, Statistical Programmes
2. Research Methodology: -Definition, types of research, designing a written protocol

Bio-Statistics: - Introduction, collection of data, presentation of data, Measures of Central tendency, measures of dispersion, Tests of significance, Sampling and sampling

	techniques- types, errors, bias, blind trails and calibration 3.Innovative Learning and Teaching Methodologies.
SECTION B (Periodontology)	
1. Introduction: Definition of Periodontology, Periodontics, Periodontia, Brief historical background, Scope of Periodontics 2. Development of perio-dontal tissues, micro-structural anatomy and biology of periodontal tissues in detail Gingiva. Junctional epithelium in detail, Epithelial-Mesenchymal interaction, Periodontal, ligament Cementum, Alveolar bone. 3. Defensive mechanisms in the oral cavity: Role of-Epithelium,Gingival fluid, Saliva and otherdefensive mechanisms in the oral environment	
4. Age changes in periodontal structures and their significance in Geriatric dentistry	Age changes in teeth and periodontal structures and their association with periodontal diseases
5. Classification of periodontal diseases	Need for classification, Scientific basis of Classification Classification of gingival and periodontal diseases as describedin World Workshop 1989 Gingivitis: Plaque associated, ANUG, steroid hormone influenced, Medication influenced, Desquamative gingivitis, other forms of gingivitis as in nutritional deficiency, bacterial and viralinfections etc. Periodontitis: Adult periodontitis, Rapidly progressive periodontitis A and B, Juvenile periodontitis (localized, generalized and post-juvenile), Prepubertal periodontitis, Refractory periodontitis
6. Gingival diseases	Localized and generalized gingivitis, Papillary, marginal anddiffuse gingivitis Etiology, pathogenesis, Clinical signs, symptoms and management of i) Plaque associated gingivitis ii) Systemically aggravated gingivitis (sex hormones,drugs and systemic diseases) iii) ANUG iv) Desquamative gingivitis - Gingivitis associated with lichen planus, pemphigoid, pemphigus and other vesiculobullous lesions v) Allergic gingivitis vi) Infective gingivitis-Herpetic, bacterial and candidial vii) Pericoronitis viii) Gingival enlargement (classification and differential diagnosis)
7. Epidemiology of periodontal diseases	Definition of index, incidence prevalence, epidemiology, endemic, epidemic and pandemic Classification of indices(Irreversible and reversible) Deficiencies of earlier indices used in Periodontics Detailed understanding of Silness and Loe Plaque Index ,Loe and Silness Gingival Index, CPITN and CPI. Prevalence of periodontal diseases in India and other countries. Public health significance(All these topics are covered at length under community dentistry. Hence, the topics may

	be discussed briefly. However, questions may be asked from the topics for examination
8. Extension of inflammation fromgingiva	Mechanism of spread of inflammation from gingival area to 1 deeper periodontal structures Factors that modify the spread
9. Pocket	Definition, signs and symptoms, classification, pathogenesis, histopathology, root surface changes and contents of the pocket
10. Etiology	- Dental Plaque (Biofilm) - Definition, New concept of biofilm - Types, composition, bacterial colonization, growth,maturation and disclosing agents - Role of dental plaque in periodontal diseases - Plaque microorganisms in detail and bacteria associated with periodontal diseases - Plaque retentive factors - Materia alba - Food debris - Calculus - Definition - Types, composition, attachment, theories of formation - Role of calculus in disease Food Impaction - Definition - Types, Etiology - Hirschfelds' classification - Signs, symptoms and sequelae of treatment Trauma from occlusion - Definition, Types - Histopathological changes - Role in periodontal disease - Measures of management in brief Habits - Their periodontal significance - Bruxism and parafunctional habits, tongue thrusting ,lip biting, occupational habits IATROGENIC FACTORS Conservative Dentistry - Restorations - Contact point, marginal ridge, surface roughness, overhanging restorations, interface between restoration and teeth Prosthodontics - Interrelationship - Bridges and other prosthesis, pontics (types), surface contour, relationships of margins to the periodontium, Gingival protection theory, muscle action theory and theory of access to oral hygiene. Orthodontics - Interrelationship, removable appliances and fixed appliances - Retention of plaque, bacterial changes Systemic diseases - Diabetes, sex hormones, nutrition(Vit.C and proteins)

	- AIDS and periodontium - Hemorrhagic diseases, Leukemia, clotting factor disorders, PMN disorders
11. Risk factors	Definition. Risk factors for periodontal diseases
12. Host response	Mechanism of initiation and progression of periodontal diseases Basic concepts about cells, Mast cells, neutrophils, macrophages, lymphocytes, immunoglobulins, complement system, immune mechanisms and cytokines in brief Stages in gingivitis-Initial, early, established and advanced Periodontal disease activity, continuous paradigm, random burst and asynchronous multiple burst hypothesis
13. Periodontitis	Etiology, histopathology, clinical signs and symptoms, diagnosis and treatment of adult periodontitis Periodontal abscess; definition,classification, pathogenesis,differential diagnosis and treatment Furcation involvement,Glickmans'classification, prognosis and management Rapidly progressive periodontitis Juvenile periodontitis: Localized and generalized Post-juvenile periodontitis Periodontitis associated with systemic diseases Refractory periodontitis
14. Diagnosis	Routine procedures, methods of probing, types of probes, (According to case history) Halitosis: Etiology and treatment. Mention advanced diagnostic aids and their role in brief
15. Prognosis	Definition, types, purpose and factors to be taken into consideration
16. Treatment plan	Factors to be considered
17. Periodontal therapy	A. General principles of periodontal therapy. Phase I,II, III, IV therapy. Definition of periodontal regeneration, repair, new attachment and reattachment. B. Plaque control i. Mechanical tooth brushes, interdental cleaning aids, dentifrices ii. Chemical; classification and mechanism of action of each and pocket irrigation
18. Pocket eradication procedures	Scaling and root planing: Indications Aims and objectives Healing following root planning Hand instruments, sonic, ultrasonic and piezo-electric scalers Curettage and present concepts Definition Indications Aims and objectives Procedures and healing response Flap surgery Definition Types of flaps, Design of flaps, papilla preservation

	Indications and contraindications - Armamentarium Surgical procedure and healingresponse
19. Osseous Surgery	Osseous defects in periodontal disease Definition Classification Surgery: resective, additive osseous surgery(osseous grafts with classification of grafts) Healing responses Other regenerative procedures; root conditioning Guided tissue regeneration
20. Mucogingival surgeryand periodontal plastic surgeries	Definition Mucogingival problems: etiology, classification of gingivalrecession (P.D.Miller Jr. and Sullivan and Atkins) Indications and objectives Gingival extension procedures: lateral pedicle graft, frenectomy, frenotomy Crown lengthening procedures Periodontal microsurgery in brief
21. Splints	Periodontal splints Purpose and classification Principles of splinting
22. Hypersensitivity	Causes, Theories and management
23. Implants	Definition, types, scope and biomaterials used. Periodontal considerations: such as implant-bone interface, implant-gingiva interface, implant failure, peri-implantitis and management
24. Maintenance phase(SPT)	Aims, objectives and principles Importance Procedures Maintenance of implants
25. Pharmaco-therapy	Periodontal dressings Antibiotics and anti-inflammatory drugs Local drug delivery systems
26. Periodontal managementof medically compromised patients	Topics concerning periodontal management of medically compromised patients
27. Inter-disciplinary care	Pulpo-periodontal involvement Routes of spread of infection Simons' classification Management
28. Systemic effects of periodontal diseases in brief	Cardiovascular diseases, Low birth weight babies etc.
29. Infection control protocol	Sterilization and various aseptic procedures
30. Ethics	

SECTION C (Oral Pathology and Microbiology)

1. INTRODUCTION:

- A bird's eye view of the different pathological processes involving the oral cavity and oral cavity involvement in systemic diseases to be brought out. Interrelationship between General Medicine and General Surgery and Oral pathology to be emphasized.

2. Developmental disturbances of teeth, jaws and soft tissues of oral and paraoral region :

- Introduction to developmental disturbances - Hereditary, Familial mutation, Hormonal etc. causes to be highlighted.
- Developmental disturbances of teeth - Etiopathogenesis, clinical features, radiological features and histopathological features as appropriate :-
  The size, shape, number, structure and eruption of teeth and clinical significance of the anomalies to be emphasized.
- Forensic Odontology.
- Developmental disturbances of jaws - size and shape of the jaws.
- Developmental disturbances of oral and paraoral soft tissues - lip and palate - clefts, tongue, gingiva, mouth, salivary glands and face.

3. Dental Caries :

- Etiopathogenesis, microbiology, clinical features, diagnosis, histopathology, immunology, prevention of dental caries and its sequelae.

4. Pulp and Periapical Pathology and Osteomyelitis.

- Etiopathogenesis and interrelationship, clinical features, microbiology, histopathology and radiological features (as appropriate) of pulp and periapical lesions and osteomyelitis.
- Sequelae of periapical abscess - summary of space infections, systemic complications and significance.

5. Periodontal Diseases :

- Etiopathogenesis, microbiology, clinical features, histopathology and radiological features (as appropriate) of gingivitis, gingival enlargements and periodontitis. Basic immunological mechanisms of periodontal disease to be highlighted.

6. Microbial infections of oral soft tissues :

- Microbiology, defence mechanisms including immunological aspects, oral manifestations, histopathology and laboratory diagnosis of common bacterial, viral and fungal infections namely :- Bacterial : Tuberculosis, Syphilis, ANUG and its complications - Cancrum Oris.
  Viral : Herpes Simplex, Varicella zoster, Measles, Mumps and HIV infection. Fungal : Candidal infection. Apthous Ulcers.

7. Common non- inflammatory diseases involving the jaws :

- Etiopathogenesis, clinical features, radiological and laboratory values in diagnosis of :
  Fibrous dysplasia, Cherubism, Osteogenesis Imperfecta, Paget's disease, Cleidocranial dysplasia, Rickets, Achondroplasia, Marfan's syndrome and Down's syndrome.

8. Diseases of TM Joint :

- Ankylosis, summary of different types of arthritis and other developmental malformations, traumatic injuries and myofascial pain dysfunction syndrome.

9. Cysts of the Oral and Paraoral region :

- Classification, etiopathogenesis, clinical features, histopathology, laboratory and radiological features (as appropriate) of Odontogenic cysts, Non-Odontogenic cysts, Pseudocysts of jaws and soft tissue cysts of oral and paraoral region.

10. Tumours of the Oral Cavity :

- Classification of Odontogenic, Non-Odontogenic and Salivary Gland Tumours. Etiopathogenesis, clinical features, histopathology, radiological features and laboratory diagnosis (as appropriate) of the following common tumours :-
  a) Odontogenic - all lesions.
  b) Non-odontogenic

- Benign Epithelial - Papilloma, Keratoacanthoma and Naevi.
- Benign Mesenchymal - Fibroma, Aggressive fibrous lesions, Lipoma, Haemangioma,Lymphangioma, Neurofibroma,Schwannoma, Chondroma, Osteoma and Tori.
- Malignant Epithelial - Basal Cell Carcinoma, Verrucous Carcinoma, Squamous Cell carcinoma and Malignant Melanoma
- Malignant Mesenchymal - Fibrosarcoma, Osteosarcoma, Giant cell tumour, Chondrosarcoma, Angiosarcoma, Kaposi's sarcoma, Lymphomas, Ewing's sarcoma and Other Reticuloendothelial tumours.

c) Salivary Gland

- Benign Epithelial neoplasms - Pleomorphic Adenoma, Warthin's tumour and Oncocytoma.

Malignant Epithelial neoplasms – Adenoid Cystic Carcinoma, Mucoepidermoid Carcinoma, Acinic Cell Carcinoma and Adenocarcinomas.

d) Tumours of Disputed Origin - Congenital Epulis and Granular Cell Myoblastoma.

e) Metastatic tumours - Tumors metastasizing to and from oral cavity and the routes of metastasis.

11. Traumatic, Reactive and Regressive lesions of Oral Cavity :

- Pyogenic and Giant cell granuloma, exostoses Fibrous Hyperplasia, Traumatic Ulcer and TraumaticNeuroma.
- Attrition, Abrasion, Erosion, Bruxism, Hypercementosis, Dentinal changes, Pulp calcifications and Resorption of teeth.
- Radiation effects of oral cavity, summary of Physical and Chemical injuries including allergic reactionsof the oral cavity.
- Healing of Oral wounds and complications - Dry socket.

12. Non neoplastic Salivary Gland Diseases :

- Sialolithiasis, Sialosis, Sialadenitis, Xerostomia and Ptyalism.

13. Systemic Diseases involving Oral cavity :

- Brief review and oral manifestations, diagnosis and significance of common Blood, Nutritional, Hormonaland Metabolic diseases of Oral cavity.

14. Mucocutaneous Lesions :

- Etiopathogenesis, clinical features and histopathology of the following common lesions. Lichen Planus, Lupus Erythematosus, Pemphigus and Pemphigoid lesions, Erythema Multiforme,Psoriasis, Scleroderma, Ectodermal Dysplasia, Epidermolysis bullosa and White sponge nevus..

15. Diseases of the Nerves :

- Facial neuralgias - Trigeminal and Glossopharyngeal. VII nerve paralysis, Causalgia.
- Psychogenic facial pain and Burning mouth syndrome.

16. Pigmentation of Oral and Paraoral region and Discolouration of teeth :

- causes and clinical manifestations.

17. Diseases of Maxillary Sinus :

- Traumatic injuries to sinus, Sinusitis, Cysts and Tumours involving antrum.

18. a) ORAL PRECANCER – CANCER; Epidemiology, aetiology, clinical and histopatholotgical features,TNM classification. Recent advances in diagnosis, management and prevention.

b) Biopsy : Types of biopsy, value of biopsy, cytology, histo chemistry and frozen sections indiagnosis of oral diseases.

19. Principles of Basic Forensic Odontology (Pre-clinical Forensic Odontology):

- Introduction, definition, aims and scope.
- Sex and ethnic (racial) differences in tooth morphology and histological age estimation
- Determination of sex and blood groups from buccal mucosa / saliva.
- Dental DNA methods
- Bite marks, rugae patterns and lip prints.
- Dental importance of poisons and corrosives.
- Overview of forensic medicine and toxicology

SECTION D (Oral and Maxillofacial Surgery)

1. Introduction, definition, scope, aims and objectives.
2. Diagnosis in oral surgery:
   (A) History taking
   (B) Clinical examination
   (C) Investigations.
3. Principles of infection control and cross-infection control with particular reference to HIV/AIDS and Hepatitis.
4. Principles of Oral Surgery -
   a) Asepsis: Definition, measures to prevent introduction of infection during surgery.
      1. Preparation of the patient
      2. Measures to be taken by operator
      3. Sterilisation of instruments - various methods of sterilisation etc.
      4. Surgery set up.
   b) Painless Surgery:
      1. Pre-anaesthetic considerations. Pre-medication: purpose, drugs used
      2. Anaesthetic considerations -
         a) Local b) Local with IV sedations
      3. Use of general anaesthetic
   c) Access:
      Intra-oral: Mucoperiosteal flaps, principles, commonly used intra oral incisions. Bone Removal: Methods of bone removal.
      Use of Burs: Advantages and precautions
      Bone cutting instruments: Principles of using chisel and osteotome.
      Extra-oral: Skin incisions - principles, various extra-oral incision to expose facial skeleton.
         a) Submandibular
         b) Pre auricular
         c) Incision to expose maxilla and orbit
         d) Bicoronal incision
   d) Control of haemorrhage during surgeryNormal Haemostasis
      Local measures available to control bleedingHypotensive anaesthesia etc.
   e) Drainage and Debridement
      Purpose of drainage in surgical wounds
      Types of drains used
      Debridement: purpose, soft tissue and bone debridement.
   f) Closure of wounds
      Suturing: Principles, suture material, classification, body response to various materials etc.
   g) Post operative care
      Post operative instructions Physiology of cold and heat Control of pain - analgesics Control of infection - antibiotics

Control of swelling - anti-inflammatory drugs Long term post operative follow up - significance.

5. Exodontia: General considerations Ideal Extraction.
   Indications for extraction of teeth
   Extractions in medically compromised patients. Methods of extraction -
   (a) Forceps or intra-alveolar or closed method. Principles, types of movement, force etc.
   (b) Trans-alveolar, surgical or open method, Indications, surgical procedure.
   Dental elevators: uses, classification, principles in the use of elevators, commonly used elevators.
   Complications of Exodontia - Complications during exodontia Common to both maxilla and mandible. Post-operative complications
   Prevention and management of complications.
6. Impacted teeth:
   Incidence, definition, aetiology.
   (a) Impacted mandibular third molar. Classification, reasons for removal, Assessment - both clinical and radiological Surgical procedures for removal. Complications during and after removal, Prevention and management.
   (b) Maxillary third molar,
   Indications for removal, classification, Surgical procedure for removal.
   (c) Impacted maxillary canine Reasons for canine impaction,
   Localization, indications for removal,
   Methods of management, labial and palatal approach, Surgical exposure, transplantation, removal etc.
7. Pre-prosthetic Surgery:
   Definition, classification of procedures
   (a) Corrective procedures: Alveoloplasty, Reduction of maxillary tuberosities, Frenoctemies and removal of tori.
   (b) Ridge extension or Sulcus extension procedures Indications and various surgical procedures
   (c) Ridge augmentation and reconstruction. Indications, use of bone grafts, Hydroxyapatite Implants - concept of osseo integration Knowledge of various types of implants and surgical procedure to place implants.
8. Diseases of the maxillary sinus Surgical anatomy of the sinus. Sinusitis both acute and chronic
   Surgical approach of sinus - Caldwell-Luc procedure Removal of root from the sinus.
   Oro-antral fistula - aetiology, clinical features and various surgical methods for closure.
9. Disorders of T.M. Joint
   Applied surgical anatomy of the joint.
   Dislocation - Types, aetiology, clinical features and management. Ankylosis - Definition, aetiology, clinical features and management
   Myo-facial pain dysfunction syndrome, aetiology, clinical features, management- Non surgical and surgical.
   Internal derangement of the joint. Arthritis of T.M. Joint.
10. Infections of the Oral cavity
    Introduction, factors responsible for infection, course of odontogenic infections, spread of odontogenic infections through various facial spaces. Dento-alveolar abscess - aetiology, clinical features and management.
    Osteomyelitis of the jaws - definition, aetiology, pre-disposing factors, classification, clinical features and management.
    Ludwigs angina - definition, aetiology, clinical features, management and complications.
11. Benign cystic lesions of the jaws - Definition, classification, pathogenesis.

Diagnosis - Clinical features, radiological, aspiration biopsy, use of contrast media and histopathology.

Management - Types of surgical procedures, Rationale of the techniques, indications, procedures, complications etc.

12. Tumours of the Oral cavity -General considerations

Non odontogenetic benign tumours occuring in oral cavity - fibroma, papilloma, lipoma, ossifying fibroma, mynoma etc.

Ameloblastoma - Clinical features, radiological appearance and methods of management.

Carcinoma of the oral cavity -Biopsy - types

TNM classification.

Outline of management of squamous

Cell carcinoma: surgery, radiation and chemotherapy

Role of dental surgeons in the prevention and early detection of oral cancer.

13. Fractures of the jaws -

General considerations, types of fractures, aetiology, clinical features and general principles of management.

mandibular fractures - Applied anatomy, classification. Diagnosis - Clinical and radiological

Management - Reduction closed and openFixation and immobilisation methods

Outline of rigid and semi-rigid internal fixation.

Fractures of the condyle - aetiology, classification, clinical features, principles of management.

Fractures of the middle third of the face.

Definition of the mid face, applied surgical anatomy, classification, clinical features and outline of management.

Alveolar fractures - methods of managementFractures of the Zygomatic complex

Classification, clinical features, indications for treatment, various methods of reduction and fixation.

Complications of fractures - delayed union, non-union and malunion.

14. Salivary gland diseases -

Diagnosis of salivary gland diseases' Sialography, contrast media, procedure. Infections of the salivary glands

Sialolithiasis - Sub mandibular duct and gland and parotid duct. Clinical features, management.

Salivary fistulae

Common tumours of salivary glands like Pleomorphic adenoma including minor salivary glands.

15. Jaw deformities -

Basic forms - Prognathism, Retrognathism and open bite.Reasons for correction.

Outline of surgical methods carried out on mandible and maxilla.

16. Neurological disorders -

Trigeminal neuralgia - definition, aetiology, clinical features and methods of management including surgical.

Facial paralysis - Aetiology, clinical features. Nerve injuries - Classification, neurorrhaphy etc.

17. Cleft Lip and Palate -

Aetiology of the clefts, incidence, classification, role of dental surgeon in the management of cleft patients. Outline of the closure procedures.

18. Medical Emergencies in dental practice –

Primary care of medical emergencies in dental practice particularly -

(a) Cardio vascular (b) Respiratory (c) Endocrine (d)Anaphylactic reaction (e) Epilepsy (f) Epilepsy

19. Emergency drugs and Intra muscular I.V. Injections -
    Applied anatomy, Ideal location for giving these injections, techniques etc.
20. Oral Implantology
21. Ethics

LOCAL ANAESTHESIA:

Introduction, concept of L.A., classification of local anaesthetic agents, ideal requirements, mode of action, types of local anaesthesia, complications.

Use of Vaso constrictors in local anaesthetic solution - Advantages, contra-indications, various vaso constrictors used.Anaesthesia of the mandible -

Pterygomandibular space - boundaries, contents etc. Interior Dental Nerve Block - various techniques Complications

Mental foramen nerve blockAnaesthesia of Maxilla - Intra - orbital nerve block.

Posterior superior alveolar nerve blockMaxillary nerve block - techniques.

GENERAL ANAESTHESIA –

Concept of general anaesthesia.

Indications of general anaesthesia in dentistry.Pre-anaesthetic evaluation of the patient.

Pre-anaesthetic medication - advantages, drugs used. Commonly used anaesthetic agents.

Complication during and after G.A.

I.V. sedation with Diazepam and Medozolam. Indications, mode of action, technique etc. Cardiopulmonary resuscitation

Use of oxygen and emergency drugs.Tracheostomy.

Note: The above syllabus is as per the B.D.S. course regulation notified by the Dental Council of India.

दिनांक – १९ मार्च, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग